AI model of a photovoltaic installation

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Abstract: This work aims to develop two models to simulate the operation of a photovoltaic system and predict electricity production using weather data, including solar radiation and system operating temperature. A prototype measurement system was developed to collect data from an operating photovoltaic installation. The obtained data set was prepared for developing linear regression and LSTM models simulating the operation of the installation. The models can be applied to estimate the value of electricity production for a specific installation in the future and the resulting ability to plan the production and the redistribution of this energy.

Keywords: : photovoltaic system modeling, LSTM, neural network, linear regression.

1 Introduction

This work describes the process of developing artificial intelligence models that simulate the operation of a photovoltaic panel system depending on the seasonal data (such as solar radiation and system operating temperature) and can be used for predicting electricity production. The problem is quite new, but there are some works that show similar models using neural or agent approaches, but rather taking into account economic aspects [Ag22], [DD26], and no one has tried to describe collecting data using a device designed for this purpose. Our approach presented in this paper focuses on presenting the task from scratch, from constructing the data collection equipment, through data processing, to creating models using selected methods. To build such a model, a special device has been constructed, which can store data directly from PV and can observe weather conditions. While there are computational models and formulas (shown in the next section) that describe PV panel parameters depending on changes in weather and other parameters, they are difficult to use directly because the specific parameters of the panels used are often unknown, as is the impact of their installation method and real weather conditions on their values. Therefore, their experimental identification is necessary. The next step was to collect the data, then process it, and use it to build models. Two models were used in this work: one simple, based on the least squares method, and the other, a much more complex one, employing an LSTM neural network. By creating two models, it was possible to compare their performance as predictors of electricity production.

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2 Basic concepts

The operation of a photovoltaic cell is based on the photovoltaic effect, a key phenomenon in solid-state physics that occurs in semiconductors (see [DLM24] or [CO17]). The basis of photovoltaic phenomena is a semiconductor material, most commonly crystalline silicon (Si). In pure silicon, the valence band is filled, and the conduction band is empty. The energy required for an electron to jump between the bands is determined by the band gap width (E_g). To facilitate conductivity, such a semiconductor is doped. To achieve n-type conductivity (electron conductivity, negative charges), it is doped with elements from group V of the periodic table, such as phosphorus (P). P-type conductivity (hole conductivity, positive charges) is obtained by doping with elements from group III, such as boron (B). At the boundary of the n and p layers, a layer of space charge is formed - the p-n junction - which is crucial for the operation of the photovoltaic cell. The photovoltaic effect is a process in which light is directly converted into electrical energy. To be absorbed and generate a carrier pair, a photon must have an energy greater than or equal to the band gap width (E_g). For crystalline silicon, $E_g \approx 1.12$ eV. Absorption leads to the transition of an electron from the valence band to the conduction band. The electric field in the p-n junction acts as a barrier. Immediately after generation, electron-hole pairs are separated. It is this process of creating and separating charges of different signs at opposite ends of the cell that generates the voltage. Total solar power at the average distance from Earth to the Sun, often described as AM0 (Air Mass Zero) power, is approximately 1361 W/m^2 (the so-called solar constant). However, after passing through the Earth's atmosphere, this radiation weakens, and at the Earth's surface it ranges from about 1200 W/m^2 at the equator, about 900 W/m^2 in Central Europe, and about 500-700 W/m^2 at the poles (at the highest position of the Sun). These parameters are called AM1.5 (Air Mass 1.5) values with the spectral composition of light appropriately modified by the atmosphere. Irradiance (G) is a fundamental indicator of the solar power reaching the module. Short-circuit current (I_{sc}) is almost perfectly linearly proportional to irradiance. This relationship is described by the formula (1):

$$I_{sc}(G) \approx \frac{I_{sc_{STC}} * G}{G_{STC}} \qquad (1)$$

An increase in irradiance causes a logarithmic increase in the open-circuit voltage (V_{oc}), which is a weaker relationship than that of current.

$$V_{OC} = \frac{n*k*T}{q} * ln\left(\frac{I_{sc}}{I_o} + 1\right) \qquad (2)$$

G, G_{STC} – irradiance, irradiance at Standard Test Conditions (STC, T=25°C, P_{light}=1000W/m^2, AM1.5 light spectrum), I_{sc}, I_{sc_STC} – short-circuit current, short-circuit current at STC, I_o - saturation current of the p-n junction, q - the charge of an electron ($1.602 \cdot 10^{-19}$ C), V_{oc} – open circuit voltage, n - diode quality factor (for Si cells ranges from 1 to 2), k - Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), T - cell temperature in Kelvin.

The power (P) of the electric current delivered at a certain moment is the product of the current and voltage (3), and the electric energy (E) is the integral of the current power during the operation of the panel (4).

$$P(t) = I(t) * V(t) \qquad (3)$$

$$E = \int_{t=t_1}^{t_2} P(t)dt \qquad (4)$$

$I(t)$ - instantaneous value of electric current, $V(t)$ - instantaneous value of electric voltage, t, t_1, t_2 – time, start and end time of the system operation.

Instantaneous values of I and V are not exactly equal to those calculated from formulas (1) and (2); those given above are rather their upper estimates. The panels produce direct current (DC) with variable parameters. Connecting them to the power grid requires an inverter that converts the DC to alternating current (AC) (see Fig. 1).

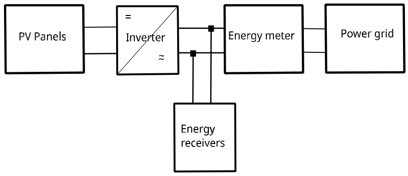


Fig. 1: Diagram of connecting PV panels and an inverter to the power grid.

Generally speaking, the values of t_1 and t_2 are determined by sunrise and sunset, but in reality, by the switching on and off of the inverter. All the above formulas show the most important factors that influence the power of the panel, such as irradiance, length of day, height of the sun above the horizon, junction temperature, and the material from which it is made, and more unpredictable factors such as cloud cover, dust, snow, hoarfrost, rain, and similar. It is not possible to examine their overall impact analytically, and experimental research is needed.

3 The system used to collect data during PV operation

Collecting data on the performance of PV panels requires a significant amount of data. Some of this data can be collected from the PV-inverter system. But data such as irradiance and panel temperature must be collected separately using a specially designed device, resembling a weather station, placed next to the panels.

3.1 The hardware

The device consists of a measurement system controller and connected light (irradiation) and temperature sensors. Access to the inverter is via a Wi-Fi router, which provides access to the local LAN and internet services (see Fig. 2). The main device of the system is a well-supported Raspberry PI minicomputer, which serves as the measurement system controller. This device is quick and easy to configure thanks to prebuilt drivers. It supports both C and Python compilers. Both languages have the necessary libraries for communication via serial interfaces, such as I2C and 1-Wire. The device is connected to sensors via GPIO ports and their associated interfaces, as well as a built-in Wi-Fi module for communication with a router and LAN. The minicomputer monitors the operation of temperature and light sensors, collecting data at predefined intervals, and reads production results from the inverter. The inverter transforms the output voltage and also monitors the operation of the photovoltaic system. Communication with the inverter is via TCP/IP using a local router, to which both the controller and the inverter are connected. The local router also provides internet connectivity for remote services such as Gmail, GitHub, and SSH.

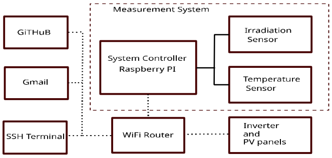


Fig. 2: System structure diagram

The measurement system reads data from the sensors, collects a 24-hour data set, which the controller saves to a CSV file kept locally on the SD card. The file is closed at midnight, and a daily report is generated and sent to Gmail. Then, the measurement system restarts the controller, logs into the LAN via Wi-Fi, and updates the software from GitHub. Next, it automatically launches the measurement script, and another daily cycle is executed.

Device[5]	Type	Bus	Parameters
Termometer	DS18B20	1-WIRE	-55°C – 125°C
Light sensor[6]	B-LUX-V30	I2C	0 – 200 klm
Microcomputer	Raspberry Pi Zero 2W WH 512MB RAM - WiFi	WiFi, Bluetooth, 1-WIRE, I2C	
Inverter	FRONIUS SYMO 6.0-3-M	WiFi	P_{max_out}[7] = 6kW
Photovoltaic modules (22pcs.)	Q.ANTUM Q.PEAK DUO-G5 305-330		P_{MPP} [8]= 330W

Tab. 1: Devices from which the system is built

5 Detailed information about the devices used can be found, in [Al26], [Df26], [Ra26], [Fr26], and [Mp26].

6 The light sensor used measures illuminance in [lm], but the energy of the light is connected with irradiance [W/m^2]. However, these quantities are linearly related for a given light source (solar light spectrum) and can be recalculated using a formula: 685 Lux [lm/m^2] = 1 W/m^2 or 1 Lux = 0.0079 W/m^2.

7 P_{max_out} - maximum inverter power fed into the power grid.

8 P_{MPP} – maximum electrical power generated under STC conditions (Maximum Power Point).

3.2 The software

The measurement system software consists of a set of scripts run in a planned order, placed together with libraries on the Raspberry PI device. The project has been placed in a GITHUB repository. You can download its latest version from the PVLogger.git repository. This solution provides great flexibility in the development and creation of code corrections. The measurement system has been programmed mainly in Python, C++, and shell commands to carry out individual subroutines in a certain logical order, which enables the implementation of the daily measurement cycle. Starting at midnight, the system executes scripts from MYSCRIPT.SERVICE containing scripts controlling the measurement system. The script AUTOSTART starts the internet connection, updates the software, and runs the main program. PVLOGGER is the main program that collects data in a loop at a specified time interval. Finally, each working day, the program sends the results and status notifications by email. Then, at midnight, it restarts and clears the data again. The block diagram of the system operation is shown in Fig. 3.

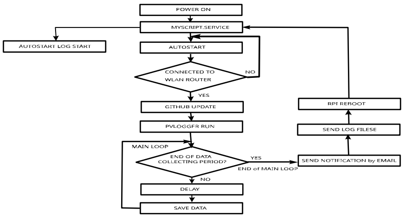
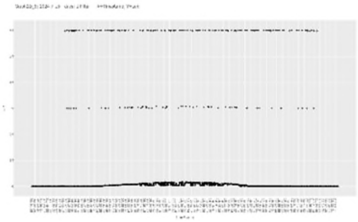


Fig. 3: The block diagram of the automatic measurement system operation

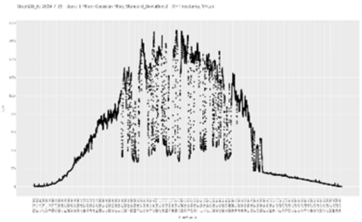
4 Results obtained

4.1 The data preprocessing

The first stage of processing the data obtained from the constructed device was to appropriately filter the data to eliminate sensor errors, interference, and data obtained without lighting. Raw data collected from the sensor during one day (2024/07/15) are presented in Fig. 4a). Apart from the data series, one can notice two series that are certainly erroneous because they contain data with impossible illuminance values. Therefore, the following data preprocessing was performed (results are shown in Fig. 4b)). The erroneous data were deleted, and the remaining data were filtered to remove noise and outliers using a Gaussian filter (5) (see [OS094]).



a)



b)

Fig. 4: a) The noisy raw data obtained from the light sensor and b) the data after preprocessing using a Gaussian filter (σ=2)

$$G(x)=\frac{1}{\sqrt{2\pi\sigma^2}}exp\left(\frac{-x^2}{2\sigma^2}\right) \tag{5}$$

$G(x)$ – the value of the Gaussian function at point x, σ – standard deviation (controls the "blurring" of the filter), x – distance from the center (can be, for example, -1, 0, 1 for a 3-element filter).

4.2 The models

Linear regression model. The method of least squares (OLS – Ordinary Least Squares) [Co03] consists of fitting a simple regression line/surface to minimize the sum of squares of differences between the observed and predicted values of the dependent variable. The coefficients of the regression surface (6) are determined from the formula (7).

$$y=a_1 * x_1+a_2 * x_2+b \tag{6}$$

$$A=(X' * X)^{-1} * X' * Y \tag{7}$$

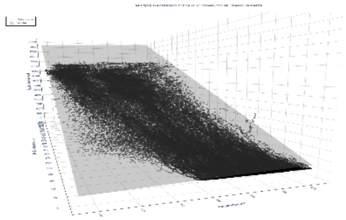
Y is a dependent data vector, and X is a matrix of independent data supplemented with a unit vector, A = [a_1, a_2, b] – a vector of searched regression parameters.

Time	a_1 (temperature)	a_2 (irradiation)	b	R^2	Number of days
07-2024	10.282	0.061775	-2942.5	0.9637	31
08-2024	14.243	0.062507	-4109.9	0.9571	12
09-2024	15.437	0.077128	-4444.7	0.9509	12
10-2024	4.7887	0.071981	-1332.1	0.9117	12
11-2024	57.765	0.050808	-1594.9	0.7200	6
12-2024	5.7580	0.077986	-1605.7	0.9424	2

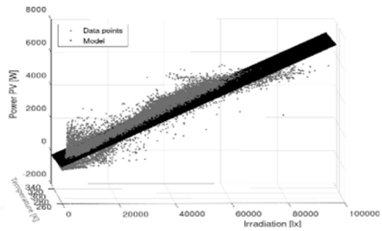
2024	3.5798	0.065990	-947.85	0.94643	78
01-2025	1.1715	0.035135	-341.31	0.3700	13
02-2025	4.9480	0.076804	-1401.0	0.9325	17
03-2025	5.7610	0.083579	-1565.0	0.9731	24
04-2025	4.1314	0.082172	-1084.8	0.9502	30
2025	5.6454	0.083158	-1566.3	0.9622	85
Total	-4.3980	0.075880	1348.1	0.9381	163

Tab. 2: Summary of linear regression results for individual months, parts of 2024 and 2025, and for all data (a_1, a_2, and b – regression parameters, R^2 - the coefficient of determination)

The model parameters presented in Tab. 2 are quite surprising, because while for all shorter periods they are quite similar, especially the dependence on irradiance, for the entire data set, the obtained model is different and, interestingly, more similar to the physical dependence of panel efficiency on temperature. Fig. 5 shows graphs presenting the features of selected models.



a)



b)

Fig. 5: 3D linear regression model obtained for all data, the dependence of electricity production a) on panel temperature; b) on solar radiation and temperature of panels

LSTM neural network. The LSTM (Long Short-Term Memory) neural network (see Fig. 6) belongs to the category of recurrent networks, which are characterized by the ability to process data in a time-dependent manner based on current and past values ([GSC00], [CO17]). This enables deeper analysis of dependencies and introduces an element of memory or inertia, which is clearly present in the operation of PV panels. The operation of the LSTM network is described by formulas (8)–(11).

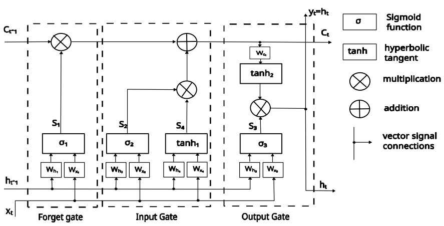


Fig. 6: Schematic diagram of the structure of one segment of the LSTM network

$$s_i=\sigma_i\left(W_{x_i}\cdot x_t+W_{h_i}\cdot h_{t-1}+b_i\right) i=1,2,3. \quad (8)$$

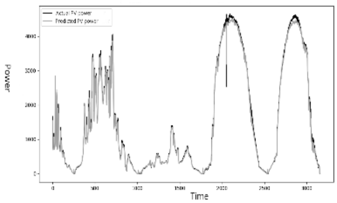
$$s_4=tanh_1\left(W_{x_4}\cdot x_t+W_{h_4}\cdot h_{t-1}+b_4\right) \quad (9)$$

$$c_t=s_1\cdot c_{t-1}+s_2\cdot s_3 \quad (10)$$

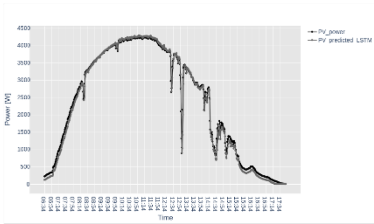
$$h_t=y_t=s_4\cdot tanh_2\left(W_{c_t}\cdot c_t+b_{c_t}\right) \quad (11)$$

$W_{x}i$, $W_{x}4$ – weights for the input signal; $W_{h}i$, $W_{h}4$, $W_{c}t$ – weights for the short-term and long-term signal; b_i, b_4, $b_{c}t$ – threshold values (biases); s_i, s_4 – auxiliary symbols for the output signals from the corresponding activation function modules; c_t - long-term signal; h_t - short-term input signal; t, t-1 – subsequent time moments.

The input values were: irradiance, panel temperature, time, day of the week, and day of the year. The vector of standard output signals was the PV power production. The data was indexed at a time resolution of 1 minute. The data was appropriately preprocessed and normalized, and divided into training and validation parts. The data selected for learning and testing the considered models were selected randomly, but using data from the whole days. The key challenge is selecting appropriate network parameters and training procedures. The well-chosen parameters yield good results with an R^2 value better than 0.99. Results obtained using the LSTM network are shown in Fig. 7. The figures show that the model obtained using the LSTM network is very accurate.



a)



b)

Fig. 7: The comparison of real and obtained results, a) real data and test data; b) LSTM predictions for March 21, 2025.

4.3 PV output power prediction

Tab. 3 presents the results of predicting the electricity production for selected days of the training data. The significant advantage of the LSTM-based model is evident, although the OLS method also gives acceptable results, considering the effort required to create the LSTM model. Tab. 4 shows the results obtained for testing data. The obtained accuracies (12) are lower than for training data, but for the LSTM, they are still very high, demonstrating the enormous potential of using it for modeling.

$$S_p=\frac{\left(E_r-\left|E_p-E_r\right|\right)*100\%}{E_r} \quad (12)$$

S_p – daily prediction accuracy, E_r – produced energy, E_p - prediction of the evaluated model.

Date	Energy [kWh]	Prediction: OLS [kWh]	Prediction: LSTM [kWh]	Accuracy (S) OLS [%]	Accuracy (S) LSTM [%]
15-08-2024	32.195	34.792	31.302	91.934	97.226
15-10-2024	5.094	5.599	4.68	90.086	91.873
15-02-2025	4.028	5.375	3.689	66.559	91.584
21-03-2025	26.621	23.231	26.026	87.266	97.765

Tab. 3: Prediction results of daily electricity production for the training data.

Date	Energy [kWh]	Prediction: OLS [kWh]	Prediction: LSTM [kWh]	Accuracy (S) OLS [%]	Accuracy (S) LSTM [%]
14-06-2025	40.911	32.546	39.769	79.355	97.209
15-07-2025	30.513	24.701	28.717	80.952	94.113
11-08-2025	29.960	24.142	28.406	80.581	94.813

Tab. 4: Prediction results of daily electricity production for the testing data.

5 Summary

The developed device allows for estimating production data from any time range and can be used as a predictor to estimate energy loss in the event of a forced shutdown. It is also possible to use the model as a predictor of energy production for a specific time period in the future. In this case, the forecast of irradiance and temperature for that time period is required. These data can be supplied either from external sources or estimated using another prediction model. It should be emphasized, that the system is quite simple, containing only two basic sensors.

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